

# PHYS 225

Week 5, Lecture 5:  
Matrix Multiplication and Taylor Series



UNIVERSITY OF  
**ILLINOIS**  
URBANA-CHAMPAIGN

# Reminders

- Homework 4 due at 11:59 pm this Thursday, 2/19.
- Office Hours **Wednesday, Thursday** (see course schedule page).
- Midterm is in 3 weeks, 3/10
  - Same general format/scope as homework, but shorter.
  - You are allowed a double-sided A4 page of your own formulas/notes, plus calculators.
  - A practice exam will be posted.
  - If you need DRES accommodations, please set them up the week before! (The night before is too late to accommodate).
- Don't forget to turn in your **1-minute papers** by 6 pm today!!

# Week 5, Lecture 5

## Matrix Multiplication and Taylor Series

### Objectives:

- Learn/review how to multiply two matrices
- Relate matrix multiplication to linear transformations
- Compute Taylor series for some common functions
- Use Taylor series to compute approximations

One Minute Paper: we often make approximations for small quantities, but “small” is a dimensionless concept. How would we define a “small” (dimensionless) velocity in relativity?

# Week 5, Lecture 5

## Matrix Multiplication

Recall from Lecture 3:

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \vec{v} = \begin{pmatrix} v^1 \\ v^2 \end{pmatrix}$$

$$M\vec{v} = \begin{pmatrix} av^1 + bv^2 \\ cv^1 + dv^2 \end{pmatrix}$$

dot product of row 1 with  $\vec{v}$   
dot product of row 2 with  $\vec{v}$

Matrix  $M$  is an “operator” which eats a vector  $\vec{v}$  and returns a vector  $M\vec{v}$

# Week 5, Lecture 5

## Matrix Multiplication

Matrix multiplication is a generalization of this:

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad N = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$$

$$MN = \begin{pmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{pmatrix}$$

entries are dot products  
of rows and columns



# Week 5, Lecture 5

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entries are dot products of rows and columns

But we know how to write dot products in index notation!

$$(MN)^i_j = M^i_k N^k_j$$

# Week 5, Lecture 5

## Matrix Multiplication

Some important facts about matrix multiplication:

$$N = \begin{pmatrix} p & q \\ r & s \end{pmatrix} \quad M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$MN = \begin{pmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{pmatrix}$$

NOT THE SAME!

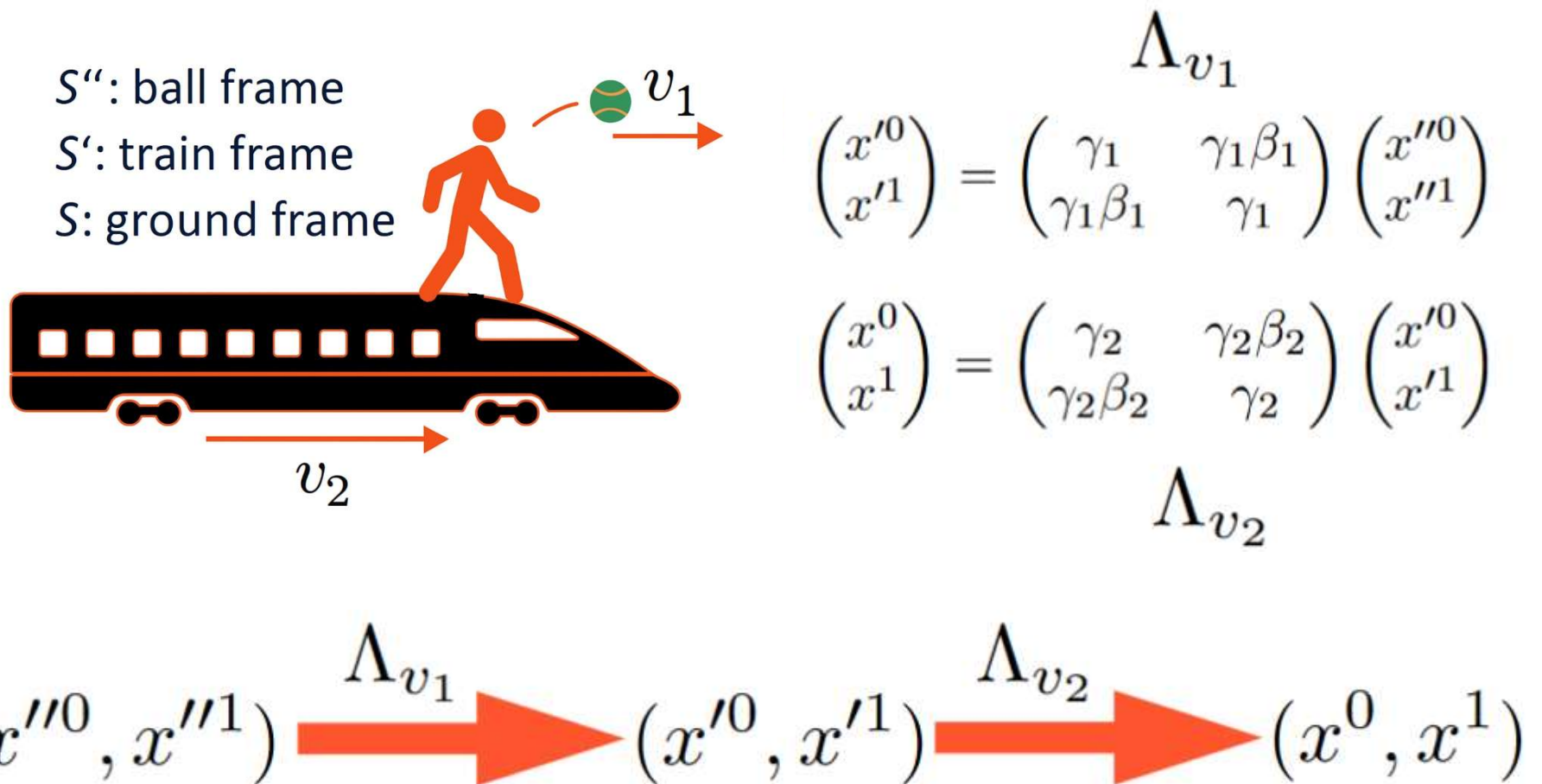
$$NM = \begin{pmatrix} ap + cq & bp + dq \\ ar + cs & br + ds \end{pmatrix}$$

Matrix multiplication  
is **not** commutative:  
order matters!

# Week 5, Lecture 5

## Matrix Multiplication

Let's use Lorentz transformations as an example:



$$\Lambda_{v_1}$$

$$\begin{pmatrix} x'^0 \\ x'^1 \end{pmatrix} = \begin{pmatrix} \gamma_1 & \gamma_1 \beta_1 \\ \gamma_1 \beta_1 & \gamma_1 \end{pmatrix} \begin{pmatrix} x''^0 \\ x''^1 \end{pmatrix}$$

$$= \begin{pmatrix} \gamma_1 x''^0 + \gamma_1 \beta_1 x''^1 \\ \gamma_1 \beta_1 x''^0 + \gamma_1 x''^1 \end{pmatrix}$$

$$\Lambda_{v_2}$$

$$\begin{pmatrix} x^0 \\ x^1 \end{pmatrix} = \begin{pmatrix} \gamma_2 & \gamma_2 \beta_2 \\ \gamma_2 \beta_2 & \gamma_2 \end{pmatrix} \begin{pmatrix} x'^0 \\ x'^1 \end{pmatrix}$$



$$\Lambda_{v_2}$$

$$\begin{pmatrix} x^0 \\ x^1 \end{pmatrix} = \begin{pmatrix} \gamma_2 & \gamma_2 \beta_2 \\ \gamma_2 \beta_2 & \gamma_2 \end{pmatrix} \begin{pmatrix} \gamma_1 x''^0 + \gamma_1 \beta_1 x''^1 \\ \gamma_1 \beta_1 x''^0 + \gamma_1 x''^1 \end{pmatrix}$$

$$= \begin{pmatrix} \gamma_2 (\gamma_1 \underline{x''^0} + \gamma_1 \beta_1 \underline{x''^1}) + \gamma_2 \beta_2 (\gamma_1 \beta_1 \underline{x''^0} + \gamma_1 \underline{x''^1}) \\ \gamma_2 \beta_2 (\gamma_1 \underline{x''^0} + \gamma_1 \beta_1 \underline{x''^1}) + \gamma_2 (\gamma_1 \beta_1 \underline{x''^0} + \gamma_1 \underline{x''^1}) \end{pmatrix}$$

$$= \begin{pmatrix} (\gamma_2 \gamma_1 + \gamma_2 \beta_2 \gamma_1 \beta_1) x''^0 + (\gamma_2 \gamma_1 \beta_1 + \gamma_2 \beta_2 \gamma_1) x''^1 \\ (\gamma_2 \beta_2 \gamma_1 + \gamma_2 \gamma_1 \beta_1) x''^0 + (\gamma_2 \beta_2 \gamma_1 \beta_1 + \gamma_2 \gamma_1) x''^1 \end{pmatrix}$$

$$= \begin{pmatrix} \gamma_2 \gamma_1 + \gamma_2 \beta_2 \gamma_1 \beta_1 & \gamma_2 \gamma_1 \beta_1 + \gamma_2 \beta_2 \gamma_1 \\ \gamma_2 \beta_2 \gamma_1 + \gamma_2 \gamma_1 \beta_1 & \gamma_2 \beta_2 \gamma_1 \beta_1 + \gamma_2 \gamma_1 \end{pmatrix} \begin{pmatrix} x''^0 \\ x''^1 \end{pmatrix}$$

$$\Lambda_{v_2}$$

$$\begin{pmatrix} x^0 \\ x^1 \end{pmatrix} = \begin{pmatrix} \gamma_2 & \gamma_2 \beta_2 \\ \gamma_2 \beta_2 & \gamma_2 \end{pmatrix} \begin{pmatrix} \gamma_1 x''^0 + \gamma_1 \beta_1 x''^1 \\ \gamma_1 \beta_1 x''^0 + \gamma_1 x''^1 \end{pmatrix}$$

$$= \begin{pmatrix} \gamma_2 \gamma_1 + \gamma_2 \beta_2 \gamma_1 \beta_1 & \gamma_2 \gamma_1 \beta_1 + \gamma_2 \beta_2 \gamma_1 \\ \gamma_2 \beta_2 \gamma_1 + \gamma_2 \gamma_1 \beta_1 & \gamma_2 \beta_2 \gamma_1 \beta_1 + \gamma_2 \gamma_1 \end{pmatrix} \begin{pmatrix} x''^0 \\ x''^1 \end{pmatrix}$$

$$\begin{pmatrix} x^0 \\ x^1 \end{pmatrix} = \begin{pmatrix} \gamma_2 & \gamma_2 \beta_2 \\ \gamma_2 \beta_2 & \gamma_2 \end{pmatrix} \begin{pmatrix} \gamma_1 & \gamma_1 \beta_1 \\ \gamma_1 \beta_1 & \gamma_1 \end{pmatrix} \begin{pmatrix} x''^0 \\ x''^1 \end{pmatrix}$$

Now consider matrix multiplication

$$\Lambda_{v_2} \Lambda_{v_1} = \begin{pmatrix} \gamma_2 & \gamma_2 \beta_2 \\ \gamma_2 \beta_2 & \gamma_2 \end{pmatrix} \begin{pmatrix} \gamma_1 & \gamma_1 \beta_1 \\ \gamma_1 \beta_1 & \gamma_1 \end{pmatrix}$$

$$= \begin{pmatrix} \gamma_2 \gamma_1 + \gamma_2 \beta_2 \gamma_1 \beta_1 & \gamma_2 \gamma_1 \beta_1 + \gamma_2 \beta_2 \gamma_1 \\ \gamma_2 \beta_2 \gamma_1 + \gamma_2 \gamma_1 \beta_1 & \gamma_2 \beta_2 \gamma_1 \beta_1 + \gamma_2 \gamma_1 \end{pmatrix}$$

Same matrix as before!

# Week 5, Lecture 5

## Matrix Multiplication

$$\begin{pmatrix} x^0 \\ x^1 \end{pmatrix} = \Lambda_{v_2} \Lambda_{v_1} \begin{pmatrix} x''^0 \\ x''^1 \end{pmatrix} \\ = \Lambda_{v_2} \left( \Lambda_{v_1} \begin{pmatrix} x''^0 \\ x''^1 \end{pmatrix} \right)$$

Now for the physics:  
you will do the algebra in HW 4  
to show that  $\Lambda_{v_2} \Lambda_{v_1}$  is a  
Lorentz transformation matrix  
with velocity...

$$\frac{\beta_1 + \beta_2}{1 + \beta_1 \beta_2} \quad \text{velocity addition!}$$

$$(x''^0, x''^1) \xrightarrow{\Lambda_{v_1}} (x'^0, x'^1) \xrightarrow{\Lambda_{v_2}} (x^0, x^1)$$

# Week 5, Lecture 5

## Matrix Multiplication

### The magic of index notation

$$\begin{pmatrix} x'^0 \\ x'^1 \end{pmatrix} = \Lambda_{v_1} \begin{pmatrix} x''^0 \\ x''^1 \end{pmatrix} \qquad x'^\mu = (\Lambda_{v_1})^\mu{}_\nu x''^\nu$$

$$\begin{pmatrix} x^0 \\ x^1 \end{pmatrix} = \Lambda_{v_2} \left( \Lambda_{v_1} \begin{pmatrix} x''^0 \\ x''^1 \end{pmatrix} \right) \qquad x^\sigma = (\Lambda_{v_2})^\sigma{}_\mu ((\Lambda_{v_1})^\mu{}_\nu x''^\nu)$$

$$(\Lambda_{v_2} \Lambda_{v_1})^\sigma{}_\nu = (\Lambda_{v_2})^\sigma{}_\mu (\Lambda_{v_1})^\mu{}_\nu \quad \text{matches our definition of matrix multiplication!}$$

Much more on this next week



# Week 5, Lecture 5

## Taylor Series

Here's a formula you probably learned in calculus:

$$f(x) = f(a) + f'(a)(x - a) + \frac{1}{2}f''(a)(x - a)^2 + \dots$$

$$= \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$$

$n^{\text{th}}$  derivative at  $x = a$ : **A NUMBER!!**

all  $x$  dependence appears as polynomials

This is the Taylor series (about  $x = a$ ) of the function  $f(x)$

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This is the Taylor series (about  $x = a$ ) of the function  $f(x)$

“Proof” (which would satisfy a physicist, but not a mathematician):

take successive derivatives of both sides and evaluate at  $x = a$ .

Two functions which have infinitely many derivatives equal “must” be the same function.

# Week 5, Lecture 5

## Taylor Series

Almost always, we evaluate at  $a = 0$ :

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

Some common examples:

# Week 5, Lecture 5

## Taylor Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

Some common examples:

$$\begin{aligned} e^x|_{x=0} &= 1 \\ \left. \frac{d}{dx} e^x \right|_{x=0} &= 1 \\ \left. \frac{d^2}{dx^2} e^x \right|_{x=0} &= 1 \\ &\vdots \end{aligned}$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

# Week 5, Lecture 5

## Taylor Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

Some common examples:

$$\begin{aligned} \sin x|_{x=0} &= 0 \\ \left. \frac{d}{dx} \sin x \right|_{x=0} &= \cos x|_{x=0} = 1 \\ \left. \frac{d^2}{dx^2} \sin x \right|_{x=0} &= -\sin x|_{x=0} = 0 \\ \left. \frac{d^3}{dx^3} \sin x \right|_{x=0} &= -\cos x|_{x=0} = -1 \\ &\vdots \end{aligned}$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$



## Week 5, Lecture 5

### Taylor Series

The practical use for Taylor series is when  $x$  is small.

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$\epsilon \ll 1 : \sin \epsilon \approx \epsilon$$

you called this  
the “small-angle”  
approximation in PHYS 211

This approximation is correct to  $\mathcal{O}(\epsilon)$  (order epsilon)

In fact, it is even correct to  $\mathcal{O}(\epsilon^2)$  because the  $x^2$  term is zero.  
The error is proportional to  $\epsilon^3$

# Week 5, Lecture 5

## Taylor Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

Some common examples:

$$\begin{array}{l} \cos x|_{x=0} = 1 \\ \left. \frac{d}{dx} \cos x \right|_{x=0} = -\sin x|_{x=0} = 0 \\ \left. \frac{d^2}{dx^2} \cos x \right|_{x=0} = -\cos x|_{x=0} = -1 \\ \left. \frac{d^3}{dx^3} \cos x \right|_{x=0} = \sin x|_{x=0} = 0 \\ \vdots \end{array} \quad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

# Week 5, Lecture 5

## Taylor Series

When the function is badly behaved at  $x = 0$ ,  
we usually change the LHS rather than the RHS:

$$\ln(0) = -\infty$$

$$\begin{aligned} \ln(1+x)|_{x=0} &= 0 \\ \frac{d}{dx} \ln(1+x) \Big|_{x=0} &= \frac{1}{1+x} \Big|_{x=0} = 1 \\ \frac{d^2}{dx^2} \ln(1+x) \Big|_{x=0} &= -\frac{1}{(1+x)^2} \Big|_{x=0} = -1 \\ &\vdots \\ \frac{d^n}{dx^n} \ln(1+x) \Big|_{x=0} &= \frac{(-1)^{n-1}(n-1)!}{(1+x)^n} \Big|_{x=0} = (-1)^{n-1}(n-1)! \end{aligned} \qquad \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

## Week 5, Lecture 5

### Taylor Series

You can use one series to get another!

$$\sin^2 x = \left( x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right)^2 = x^2 - \frac{x^4}{3} + \frac{2x^6}{45} + \dots$$

Often much more convenient than computing derivatives.

Warning: make sure you keep **all** terms up to the desired order!

$$\sin^2 x = \left( x - \frac{x^3}{3!} + \dots \right)^2 \quad \text{misses the } x^5 \text{ times } x \text{ term that gives an } x^6 \text{ contribution}$$

# Week 5, Lecture 5

## Taylor Series

A fun trick: you can even differentiate series term-by-term!

$$\begin{aligned}\frac{1}{1+x} &= \frac{d}{dx} \ln(1+x) \\ &= \frac{d}{dx} \left( x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \right) \\ &= 1 - x + x^2 - \dots\end{aligned}$$

This is probably **the** most useful Taylor series in physics

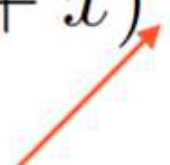


# Week 5, Lecture 5

## Taylor Series

Some extremely common variations:

$$\frac{1}{1-x} = 1 + x + x^2 + \dots$$

$$(1+x)^\alpha = 1 + \alpha x + \dots$$


binomial expansion: finite series if alpha is an integer,  
infinite series if not. We almost always only need the first term

# Week 5, Lecture 5

## Matrix Multiplication and Taylor Series

### Summary

- Matrix multiplication involves taking dot products of rows and columns
- Matrix multiplication represents the successive effects of two linear transformations — but order matters!
- Taylor series can be computed from the definition, or derived from other known series to a desired order
- Taylor series are useful for computing approximations when small quantities are involved

One Minute Paper: differentiate the first two terms of the Taylor series for  $\sin x$  to obtain the first two terms of the Taylor series for  $\cos x$

Velocity addition in natural units:

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$$V_{\text{add.}} = \frac{v_1 + v_2}{1 + \frac{v_1 v_2}{c^2}} \xrightarrow{\text{divide by } c} \frac{v_{\text{add.}}}{c} = \frac{v_1/c + v_2/c}{1 - (\frac{v_1}{c})(\frac{v_2}{c})}$$
$$\beta_{\text{add.}} = \frac{\beta_1 + \beta_2}{1 + \beta_1 \beta_2}$$

Practice with Taylor series:

For  $f(x) = \ln(1+x)$ :

$$f^{(1)}(x) = \frac{1}{1+x}$$

By chain Rule,  $f^{(2)}(x) = \frac{d}{dx} (1+x)^{-1} = -1(1+x)^{-2} \left( \frac{d}{dx} (1+x) \right)$

$$= -\frac{1}{(1+x)^2}$$

$$f^{(3)}(x) = \frac{d}{dx} (-(1+x)^{-2}) = \frac{2}{(1+x)^3}$$

So the coefficient of  $x^3$  in the Taylor series is  $\frac{f^{(3)}(0)}{3!} = \frac{2}{3!} = \frac{1}{3}$

In general, the derivatives will bring down all but the factor of  $n$  in  $n!$ , leaving a coefficient of  $\frac{1}{n}$ .

$$\sin^2 x = \left( x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right)^2 \leftarrow \text{expand to } \theta(x^6):$$
$$= x^2 + \left( \frac{-x^3}{3!} \right)^2 + \left( \frac{x^5}{5!} \right)^2 - 2(x) \left( \frac{x^3}{3!} \right) + 2(x) \left( \frac{x^5}{5!} \right) - 2 \left( \frac{x^3}{3!} \right) \left( \frac{x^5}{5!} \right)$$
$$= x^2 + \frac{x^6}{36} - \frac{x^4}{3} + \frac{x^6}{60} + \theta(x^7)$$
$$= x^2 - \frac{x^4}{3} + \frac{2x^6}{45} + \theta(x^7)$$